

MORE RESIDUALLY SOLVABLE ONE-RELATOR GROUPS

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ABSTRACT. We say a word in a free group is “partially positive” if there is at least one generator which appears in the word, but does not appear with negative exponent. We show that one-relator groups with a partially positive relator are residually solvable. This extends a theorem of Baumslag showing one-relator groups with positive relators are residually solvable.

1. INTRODUCTION

We will begin with the seminal theorem of Baumslag:

Theorem 1.1 ([1, Thm. 1]). *Let F_X be a free group. If w is a word in the “positive submonoid” of F_X , i.e. the submonoid generated by X (and not X^{-1}), then the group $G = F_X / \langle\langle w \rangle\rangle = \langle X \mid w = 1 \rangle$ is residually solvable.*

We will define residual solvability, but the internal mechanics are of less importance than the fact it is an isomorphism invariant.

Definition 1.2. A group G is residually solvable if its derived subgroups converge to the trivial group. That is,

$$\{1\} = G^{(\omega)} := \bigcap_{i=0}^{\infty} G^{(i)}$$

where $G^{(0)} := G$ and $G^{(i+1)} := [G^{(i)}, G^{(i)}]$.

What one ought to take away is that residual solvability is a condition about commutator subgroups (and so are solvability and some related notions we will explore later).

If we want to know how strong an isomorphism invariant is, a good question to ask is how probable it is that two random groups are distinguished by the isomorphism invariant. To know this we can ask what the probability that a random group is residually solvable is.

Definition 1.3. Let $\mathbb{P}_r^n(P)$ be the probability that for a uniformly sampled word w with $|w| \leq n$, $F_r / \langle\langle w \rangle\rangle$ has property P . We define $\mathbb{P}_r^\infty(P) := \lim_{n \rightarrow \infty} \mathbb{P}_r^n(P)$.

This of course sets us up for the question:

Question 1.4. *What is the probability that a random one-relator group is residually solvable?*

This question is a more general version of a question in [2] which asks simply whether $\mathbb{P}_r^\infty(\text{residual solvability}) = 1$.

Now, Theorem 1.1 tells us that positive one-relator groups are residually solvable, so it gives us a lower bound on the probability function $\mathbb{P}_r^n(\text{residually solvable})$.

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The number of positive words grows as r^n , and the total number of reduced words grows as $2r(2r-1)^n$, meaning that the probability that a random word is positive converges to zero exponentially fast (so long as the rank is at least 2, which we will generally assume from now on).

One can expand the scope of Theorem 1.1 using a trick. Applying a free group automorphism to the relator yields an isomorphic group, so it preserves isomorphism invariants. Motivated by this fact, we define potentially positive words to be those which are automorphic to positive words. Potentially positive words were introduced in the 2002 edition of [6]. In the current edition, they are addressed in problem (F33):

Question 1.5 ([6, (F33)]). *Let potential positivity be defined as above.*

- (a) *Is the property of being potentially positive algorithmically recognizable?*
- (c) *Let $P(r, n)$ denote the number of potentially positive elements of length n in F_r . What is the growth of $P(r, n)$ as a function of n , with r fixed?*

Goldstein first provided an algorithm in rank 2 in [5]. [7] and [11] improve its time complexity, ultimately giving a linear time algorithm.¹ [3] uses Goldstein's description of potentially positive elements to solve (c) for F_2 . Scant progress has been made on these problems in ranks higher than 2.

While we do not have a full description of potential positivity in ranks 3 and up, we can still make observations about this property which expand the scope of Theorem 1.1. One such observation is the following:

Observation 1.6. Any word $w \in F_r$ in which all but one generator appears positively is potentially positive.

One could hope to expand the scope even further.

Question 1.7 (Vladimir Shpilrain, personal communication). *A word $w \in F_X$ is partially positive if there is some $a \in X$ such that w can be written with only positive exponents on a , and a appears at least once in w . Is there a partially positive word which is not potentially positive?*

Restricted to rank 2 the answer is “no” per Observation 1.6, however, for higher ranks the answer seems almost certainly to be “yes”. Nevertheless, the question is open. One way we might try to resolve this is to turn Theorem 1.1 on its head. We can look for partially positive words which relate non-residually solvable groups.

However, there is an obstacle to producing such examples: they do not exist.

Theorem 1.8 (Main result). *Let F_X be a free group. If $w \in F_X$ is a partially positive word then the group $G = F_X / \langle\langle w \rangle\rangle = \langle X \mid w = 1 \rangle$ is residually solvable.*

While we fail to answer Question 1.7, we have a pretty extensive generalization of Baumslag's result.

2. METHODS

As is often the case, we will prove a slightly stronger theorem than our stated main result.

Theorem 2.1. *Let F_X be a free group and $w \in F_X$ be a partially positive word.*

¹Asymptotic complexity can be thought of as an objective measure of descriptive power. The faster or more space efficient an algorithm, the more you have said about a problem.

- If w is not a proper power, then $F_X/\langle\langle w \rangle\rangle$ is residually $\mathbb{Q}$ -solvable.
- If w is a proper power, then $F_X/\langle\langle w \rangle\rangle$ is free-by-solvable.

Residual $\mathbb{Q}$ -solvability and free-by-solvability are both generalizations of residual solvability. We will start by defining free-by-solvable groups:

Definition 2.2. A group G is free-by-solvable, if its derived series contains a free group. That is, there is some n such that $G^{(n)}$ is free.

Equivalently, G is free-by-solvable if there is a homomorphism $\varphi : G \rightarrow Q$ where Q is a solvable group and $\ker(\varphi)$ is free.

Since a free group is itself residually solvable, every free-by-solvable group is as well.

When we look at residually solvable groups, this generalization feels pretty natural. There are many groups which are almost solvable but are obstructed because they contain a non-abelian free subgroup. For example, by the Freiheitssatz, all one-relator groups with more than two generators contain non-abelian free subgroups.

Definition 2.3. A group G is residually $\mathbb{Q}$ -solvable if its rational derived series converges to the trivial group. That is,

$$\{1\} = G_{\mathbb{Q}}^{(\omega)} := \bigcap_{i=0}^{\infty} G_{\mathbb{Q}}^{(i)}$$

where $G_{\mathbb{Q}}^{(0)} := G$ and $G_{\mathbb{Q}}^{(i+1)}$ is the subgroup of all elements which have a non-zero power in $[G_{\mathbb{Q}}^{(i)}, G_{\mathbb{Q}}^{(i)}]$.

It should be clear that $G_{\mathbb{Q}}^{(\omega)} \supseteq G^{(\omega)}$; thus, residual $\mathbb{Q}$ -solvability implies residual solvability.

The thrust of this paper comes from results in [9]. In particular, the following:

Theorem 2.4 ([9, Cor. 1.2]). *Let F be a free group, $w \in F$ a word, and $G = F/\langle\langle w \rangle\rangle$ a one-relator group. The following are equivalent:*

- (1) G is residually $\mathbb{Q}$ -solvable.
- (2) $G_{\mathbb{Q}}^{(n)}$ is a free group for some integer $n \geq 0$.
- (3) For all words $r \in F$ such that $|r| \leq |w|$, and all integers $k \geq 1$, if $w \in r^k [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$, w is conjugate to r or r^{-1} .

In [9], Linton is able to use this result to create an algorithm that decides whether a one-relator group is residually $\mathbb{Q}$ -solvable by checking all possible values for r up to the length of the word.

We will also require the following lemma which is used implicitly in [9]:

Lemma 2.5. *Let F be a free group. Let $w \in F$ be a word which is not a proper power. If $F/\langle\langle w \rangle\rangle$ is residually $\mathbb{Q}$ -solvable then $F/\langle\langle w^n \rangle\rangle$ is free-by-solvable for all n .*

Proof. By the proof of [9, Lemma 4.2], there exists a solvable group Q and a homomorphism $\varphi : F/\langle\langle w^n \rangle\rangle \rightarrow Q$ such that $\ker(\varphi)$ is the free product of cyclic groups. Assume without loss of generality that φ is onto.

By the Kurosh subgroup theorem ([8, Thm. 3.01]) $[\ker(\varphi), \ker(\varphi)]$ is free. Since the commutator subgroup is a characteristic subgroup and $\ker(\varphi)$ is normal in $F/\langle\langle w^n \rangle\rangle$, $[\ker(\varphi), \ker(\varphi)]$ is a normal subgroup of $F/\langle\langle w^n \rangle\rangle$. Thus, let

$$Q' = (F/\langle\langle w^n \rangle\rangle)/[\ker(\varphi), \ker(\varphi)]$$

Since $\ker(\varphi)$ is a subgroup of $F/\langle\langle w^n \rangle\rangle$, $\text{Ab}(\ker(\varphi))$ is a subgroup of Q' . Thus, there is a natural inclusion map $\text{Ab}(\ker(\varphi)) \rightarrow Q'$.

Now we have the following short exact sequence:

$$1 \rightarrow \text{Ab}(\ker(\varphi)) \rightarrow Q' \rightarrow Q \rightarrow 1$$

Since $\text{Ab}(\ker(\varphi))$ is solvable (Abelian) and Q is solvable, Q' is the extension of a solvable group by a solvable group. Thus, Q' is solvable, and so $F/\langle\langle w^n \rangle\rangle$ is free-by-solvable. $\square$

We will also use Fox derivatives. The Fox derivative is an algebraic object which captures geometric information about commutator subgroups.

Definition 2.6 ([4]). Let F_X be a free group. The Fox derivative is a function $X \rightarrow (F_X \rightarrow \mathbb{Z}[F_X])$, written $\frac{\partial w}{\partial x}$ for $w \in F_X$ and $x \in X$. It is defined to follow the following laws:

$$\begin{aligned} (1) \quad & \frac{\partial x}{\partial x} = 1 \\ (2) \quad & \frac{\partial y}{\partial x} = 0 \quad \text{where } y \in X \setminus \{x\} \\ (3) \quad & \frac{\partial \varepsilon}{\partial x} = 0 \\ (\text{Product rule}) \quad & \frac{\partial uv}{\partial x} = \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} \end{aligned}$$

Most useful to us is a theorem of Fox:

Theorem 2.7 ([10]). Let F_X be a free group. Let $\varphi : F_X \rightarrow G$ be a group homomorphism, and let $\varphi^\dagger : \mathbb{Z}[F_X] \rightarrow \mathbb{Z}[G]$ be its natural lift to the group ring. Then for every $w \in F_X$,

$$\left(\forall x \in X : \varphi^\dagger \left(\frac{\partial w}{\partial x} \right) = 0 \right) \iff w \in [\ker(\varphi), \ker(\varphi)]$$

3. RESULTS

Lemma 3.1. Let w be a word in a free group F and a be a generator of F . Let r and $k \geq 1$ be such that:

$$(4) \quad w \in r^k [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$$

Let $\varphi_r : F \rightarrow F/\langle\langle r \rangle\rangle$ be the quotient map. Then

$$\varphi_r \left(\frac{\partial w}{\partial a} \right) = k \varphi_r \left(\frac{\partial r}{\partial a} \right)$$

Proof. Let $c \in [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$ be the solution to the equation $w = r^k c$ implied by (4). We apply the Fox derivative with respect to a , and then the quotient map φ_r (lifted

to the group ring).

$$\begin{aligned}
\varphi_r \left(\frac{\partial w}{\partial a} \right) &= \varphi_r \left(\frac{\partial r^k c}{\partial a} \right) \\
\text{(Product rule)} \quad &= \varphi_r \left(\frac{\partial r^k}{\partial a} \right) + \varphi_r(r^k) \varphi_r \left(\frac{\partial c}{\partial a} \right) \\
\text{(Theorem 2.7)} \quad &= \varphi_r \left(\frac{\partial r^k}{\partial a} \right) + 1 \cdot 0
\end{aligned}$$

Iterating the product rule of Fox derivatives, we see that

$$\begin{aligned}
\varphi_r \left(\frac{\partial r^k}{\partial a} \right) &= \varphi_r \left(\sum_{i=0}^{k-1} r^i \frac{\partial r}{\partial a} \right) \\
&= k \varphi_r \left(\frac{\partial r}{\partial a} \right)
\end{aligned}$$

□

Notation 3.2. We will use “quandle notation” for denoting group conjugation.

$$y \triangleleft x := yxy^{-1}$$

Lemma 3.3. *Let F_X be a free group and $\varphi : F_X \rightarrow G$ be a homomorphism. If $w \in \ker(\varphi)$ then*

$$\varphi \left(\frac{\partial q \triangleleft w}{\partial a} \right) = \varphi \left(q \frac{\partial w}{\partial a} \right)$$

for all $q \in F_X$ and $a \in X$.

Proof.

$$\begin{aligned}
\text{(Product rule)} \quad \varphi \left(\frac{\partial q w q^{-1}}{\partial a} \right) &= \varphi \left(\frac{\partial q}{\partial a} + q \frac{\partial w}{\partial a} + q w \frac{\partial q^{-1}}{\partial a} \right) \\
&= \varphi \left(\frac{\partial q}{\partial a} + q \frac{\partial w}{\partial a} + q \frac{\partial q^{-1}}{\partial a} \right) \\
\text{(Product rule)} \quad &= \varphi \left(\frac{\partial q q^{-1}}{\partial a} + q \frac{\partial w}{\partial a} \right) \\
(3) \quad &= \varphi \left(q \frac{\partial w}{\partial a} \right)
\end{aligned}$$

□

Lemma 3.4. *Let w be a partially positive word in a free group F . Let $r \in F$ and $k \geq 1$ be such that $w \in r^k [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$. Then r is conjugate to a word r' such that the positive generator in w appears only positively in r' .*

Proof. Again, let $\varphi_r : F \rightarrow F/\langle\langle r \rangle\rangle$ be the natural quotient map, and let a be the generator of F which appears only positively in w .

If r is not cyclically reduced, then there is some q such that $q \triangleleft r$ is cyclically reduced. Let $r' = q \triangleleft r$. Thus, $q \triangleleft w \in r'^k [\langle\langle r' \rangle\rangle, \langle\langle r' \rangle\rangle]$, for partially positive w and

cyclically reduced r' . From here we can derive the following:

$$(Lemma\ 3.1) \quad k\varphi_r\left(\frac{\partial r'}{\partial a}\right) = \varphi_r\left(\frac{\partial q \triangleleft w}{\partial a}\right)$$

$$(Lemma\ 3.3) \quad = \varphi_r\left(q\frac{\partial w}{\partial a}\right)$$

Assume that r' contains a^{-1} , and therefore can be written as a product $r_0a^{-1}r_1$, without cancellation, for some r_0 and r_1 .

$$\begin{aligned} \varphi_r\left(\frac{\partial r_0a^{-1}r_1}{\partial a}\right) &= \varphi_r\left(\frac{\partial r_0a^{-1}}{\partial a}\right) + \varphi_r\left(r_0a^{-1}\frac{\partial r_1}{\partial a}\right) \\ &= \varphi_r\left(\frac{\partial r_0}{\partial a}\right) - \varphi_r(r_0a^{-1}) + \varphi_r\left(r_0a^{-1}\frac{\partial r_1}{\partial a}\right) \end{aligned}$$

We now have a negative term, $-\varphi_r(r_0a^{-1})$. Now since $\varphi_r(q\frac{\partial w}{\partial a})$ contains only positive coefficients, and $-\varphi_r(r_0a^{-1})$ is a negative coefficient, there must be some term elsewhere which cancels it. In other words, $\varphi_r(\frac{\partial r_1}{\partial a})$ must contain a positive constant term. Thus r_1 must have a prefix of sa (without cancellation) where $\varphi_r(s) = 1$. However, s must be a proper subword of r and a classic result of Weinbaum, [12, Thm. 2] shows that no proper non-empty subword of r can be trivial under φ_r . Thus r cannot contain a^{-1} terms. $\square$

Lemma 3.5. *Let F be a free group. Let $w \in F$ be partially positive and not a proper power. If r and $k \geq 0$ are such that*

$$w \in r^k [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$$

then r is conjugate to w .

Proof. Once again, let $\varphi_r : F \rightarrow F/\langle\langle r \rangle\rangle$ be the natural quotient map, and let a be the generator of F which appears only positively in w . Let $r = q \triangleleft r'$ where r' is partially positive by Lemma 3.4 and the exponent sum of a in q , which we denote $|q|_a$, is 0 modulo $|r'|_a$. Such a q must exist for every modular residue by the following argument: Suppose we have some q with exponent sum t . Let r_0a be the shortest prefix of r' containing an a . Then

$$\begin{aligned} r &= qr'q^{-1} \\ &= qr_0aa^{-1}r_0^{-1}r'r_0aa^{-1}r_0^{-1}q^{-1} \\ &= qr_0a \triangleleft (a^{-1}r_0^{-1} \triangleleft r') \end{aligned}$$

$a^{-1}r_0^{-1} \triangleleft r'$ is partially positive, and $|qr_0a|_a = t + 1$. Then by induction, every modular residue is possible. In the same way we can stipulate that r' ends with a . If it does not, simply cycle it until it does.

Let us assume without loss of generality that w also ends in a . We will express w and r' as products:

$$\begin{aligned} w &= \prod_{i=0}^{nk-1} w_i a \\ r' &= \prod_{i=0}^{n-1} r_i a \end{aligned}$$

where $n = |r|_a$ and each w_i and r_i does not contain a (or a^{-1}). $|w|_a = nk$, because $w = r^k c$ for some $c \in [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle] \subseteq [F, F]$. Their Fox derivatives follow:

$$\begin{aligned}\frac{\partial w}{\partial a} &= \sum_{i=0}^{nk-1} w_0 \prod_{j=1}^i a w_j \\ \frac{\partial r'}{\partial a} &= \sum_{i=0}^{n-1} r_0 \prod_{j=1}^i a r_j\end{aligned}$$

For brevity, we will let the i th term of the first sum be $\hat{w}_i$, and likewise $\hat{r}_i$ is the i th term of the second sum.

We can derive the following relationship between the Fox derivatives of w and r' :

$$\begin{aligned}(\text{Lemma 3.1}) \quad \varphi_r \left(\frac{\partial w}{\partial a} \right) &= k \varphi_r \left(\frac{\partial r}{\partial a} \right) \\ &= k \varphi_r \left(\frac{\partial q \triangleleft r'}{\partial a} \right) \\ (\text{Lemma 3.3}) \quad &= k \varphi_r \left(q \frac{\partial r'}{\partial a} \right)\end{aligned}$$

Each application of the defining relation $r = 1$ on some word changes its exponent sum on a by n . Since $|\hat{w}_i|_a = |\hat{r}_i|_a = i$ and $|q|_a \equiv 0 \pmod n$, $\varphi_r(\hat{w}_i) = \varphi_r(q \hat{r}_j)$ only if $i \equiv j \pmod n$. Since there are only n terms in $\frac{\partial r'}{\partial a}$, each has a unique exponent sum of a modulo n . Thus φ_r of each term in $\frac{\partial r'}{\partial a}$ must be unique. For each $i < n$, there must be exactly k terms in $\varphi_r \left(\frac{\partial w}{\partial a} \right)$ equal to $\varphi_r(q \hat{r}_i)$. There are exactly k terms $\hat{w}_j$ where j has any particular modular residue, thus

$$\varphi_r(\hat{w}_j) = \varphi_r(q \hat{r}_i) \iff i \equiv j \pmod n$$

Now consider w_j and r_i for some $1 \leq j \leq nk-1$ and $i \equiv j \pmod n$. By definition, $\hat{w}_j = \hat{w}_{j-1} a w_j$; thus $w_j = a^{-1} \hat{w}_{j-1}^{-1} \hat{w}_j$. Similarly, $r_i = a^{-1} \hat{r}_{i-1}^{-1} \hat{r}_i$. Thus,

$$\begin{aligned}\varphi_r(w_j) &= \varphi_r(a^{-1} \hat{w}_{j-1}^{-1} \hat{w}_j) \\ &= \varphi_r(a^{-1} (q \hat{r}_{i-1})^{-1} (q \hat{r}_i)) \\ &= \varphi_r(a^{-1} \hat{r}_{i-1}^{-1} \hat{r}_i) \\ &= \varphi_r(r_i)\end{aligned}$$

Since neither w_j nor r_i contain a by definition, Freiheitssatz implies that $w_j = r_i$ when $0 \neq j \equiv i \pmod n$. It still remains to show that this is the case when $j = i = 0$. To do this we will rotate w by conjugating by $w_{nk-1} a = r_{n-1} a$. This will move w_0 into the second position, and then we will be able to use the above technique to show the desired claim. Naturally, since $w = r^k c$ for some $c \in [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$, after conjugation we have $w_{nk-1} a \triangleleft w = (r_{n-1} a \triangleleft r)^k \tilde{c}$ for some $\tilde{c} \in [\langle\langle r \rangle\rangle, \langle\langle r \rangle\rangle]$. Recall $r = q \triangleleft r'$ where r' ends with a and $|q|_a \equiv 0 \pmod{|r|_a}$. Thus, letting $\tilde{w} = w_{nk-1} a \triangleleft w$, $\tilde{r}' = r_{n-1} a \triangleleft r'$, and $\tilde{q} = r_{n-1} a \triangleleft q$,

$$\tilde{w} = (\tilde{q} \triangleleft \tilde{r}')^k \tilde{c}$$

This satisfies all the necessary conditions. Specifically:

- $\tilde{w} \in (\tilde{q} \triangleleft \tilde{r}')^k [\langle\langle \tilde{r}' \rangle\rangle, \langle\langle \tilde{r}' \rangle\rangle]$.
- $\tilde{w}$ is partially positive, since it is a rotation of w .

- $\tilde{w}$ ends with a .
- $\tilde{r}'$ is partially positive, since it is a rotation of r' .
- $\tilde{r}'$ ends with a .
- $|\tilde{q}|_a \equiv 0 \pmod{|\tilde{r}|_a}$ since $\tilde{q}$ is conjugate to q and $\tilde{r}$ is conjugate to r and $|q|_a \equiv 0 \pmod{|r|_a}$.

With these assumptions, the proof above shows that $w_0 = r_0$.

Since w is not a proper power, $k = 1$ and $w = r'$. r' is conjugate to r so w is conjugate to r . $\square$

Now we can prove our main result.

Proof of Theorem 2.1. This follows from Lemma 3.5, Theorem 2.4, and Lemma 2.5 $\square$

4. FURTHER DIRECTIONS

We have shown that for every partially positive one-relator group G there is some n such that $G^{(n)}$ is free. We call G “free-by- n -solvable”. This is an isomorphism invariant, and it has the potential to distinguish non-isomorphic partially positive one-relator groups.

Question 4.1. *When is a partially positive one-relator group free-by- n -solvable?*

This question would be particularly compelling if partially positive words were very common. While partially positive words greatly extend Baumslag’s result on positive words, we will see shortly that the asymptotic probability that a word is partially positive is 0. That is in some sense, long partially positive words are very rare.

To show this we will introduce a concept.

Definition 4.2. The *exponential growth rate* of a set of reduced words $S \subseteq F_r$, is

$$\inf\{b \in \mathbb{R} : g(S) \in O(b^n)\}$$

where $g(S)$ is the counting function of S :

$$g(S) = n \mapsto |\{s \in S : |s| \leq n\}|$$

The growth rate is closely related to the notion of topological entropy in symbolic dynamics. Clearly if the growth rate of S is less than $2r - 1$ (the growth rate of F_r), then the probability of being in S converges to 0 exponentially.

Theorem 4.3. *The set of partially positive words of length n in the free group of rank r has an exponential growth rate of $\sqrt{r^2 - 2r + 2} + r - 1 < 2r - 1$.*

Proof. The exponential growth rate can be obtained by solving a recurrence relation. Let $a \in X_r$ be the positive generator. Since a is the only letter with a special rule all other letters behave identically. An a can follow any letter, $2r - 2$ non- a letters can follow an a , and $2r - 3$ non- a letters can follow a non- a letter (every non- a letter but its inverse). We can encode this as a matrix.

$$\begin{bmatrix} 2r - 3 & 1 \\ 2r - 2 & 1 \end{bmatrix}$$

The largest positive eigenvalue for this matrix is $\sqrt{r^2 - 2r + 2} + r - 1$ which gives us the exponential growth rate. $\square$

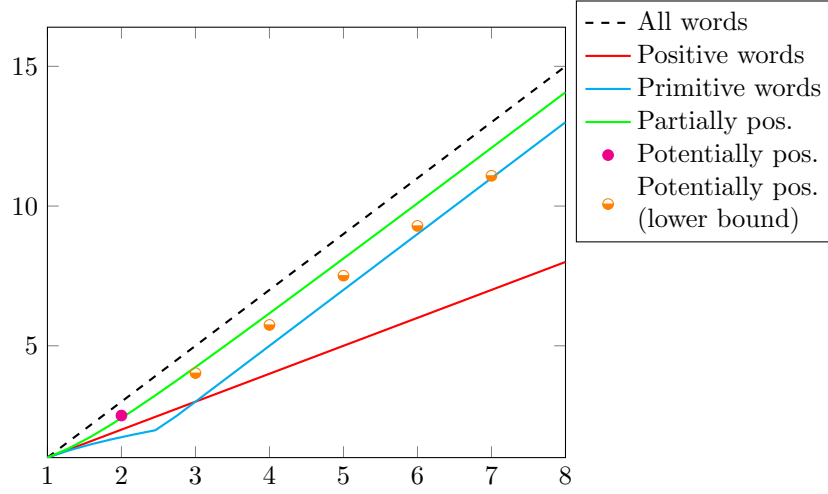


FIGURE 4.4. The exponential growth rates of various families of words which give residually solvable one-relator groups.

While partially positive words have asymptotic probability 0, $\sqrt{r^2 - 2r + 2} + r - 1$ exceeds previous lower bounds for the growth rate of potentially positive words in F_r when $r \geq 3$. To our knowledge, this constitutes an improvement on the lower bound on the exponential growth rate of words relating a residually solvable group.

Definition 4.5. We call a word in F_X potentially partially positive, or 3P, if it is automorphic to a partially positive word.

Example 4.6. The word $w = a^{-1}bac^{-1}a^2b^{-1}a^{-2}c^4 \in F(a, b, c)$ is 3P but not partially positive.

Proof. Consider the automorphism $\varphi = a \mapsto a; b \mapsto b; c \mapsto a^2ca$. $\varphi(w) = [b, c^{-1}](a^3c)^4$ which has only positive exponents on a . $\square$

Naturally, we ask the same questions of 3P as have already been asked for potentially positive words:

Question 4.7. *Is there an algorithm to decide whether an input word is 3P?*

Question 4.8. *What is the asymptotic probability that a word in F_r is 3P? What is the exponential growth rate of a 3P word in F_r ?*

Question 4.9. *Is there a “3P-blocking word”, that is a word which is not a subword of any cyclically reduced 3P word in F_r ?*

If the answer to this question is “yes”, then the asymptotic probability that a word is 3P is 0.

All of these questions have been answered for rank 2 potentially positive words, and since 3P is potential positivity in F_2 , they are settled for 3P.

Which naturally leads us back to Shpilrain’s question:

Question 4.10. *Are 3P and potentially positive the same?*

Our main result demonstrates that one cannot use Baumslag’s result to answer this question.

Both deciding 3P and potential positivity are special cases of a more general problem. Positive words form a subsemigroup of the free group, and while all partially positive words do not form a subsemigroup, if we fix a generator a and take the subset of partially positive words for which a is the positive generator we get a subsemigroup whose automorphic closure is 3P.

Problem 4.11. Given a subsemigroup $M \subseteq F_r$ and a word $w \in F_r$ determine whether w is in the automorphic closure of M , $\text{Aut}(F_r)M$.

Of course, in general this problem is hopeless. M need not be finitely generated, and indeed the subsemigroup whose automorphic closure is 3P is not finitely generated for $r \geq 2$. However, if we were able to decide the automorphic closure of the monoid K generated by $X^\pm / \{a^{-1}\}$ in F_X , it would go a long way to deciding 3P. K contains the subsemigroup whose automorphic closure is 3P, and their difference is just words which do not contain a or a^{-1} . There are efficient algorithms for deciding whether a word is automorphic to a word not containing all the generators, and such words are asymptotically negligible.

Question 4.12. *Is there a finitely generated subsemigroup $M \subseteq F_r$ such that its automorphic closure in F_r is not a decidable set?*

We conjecture that the answer is “yes”.

Question 4.13. *Is there a finitely generated subsemigroup $M \subseteq F_r$ such that $\mathbb{P}_r^\infty(M) = 0$ but $\mathbb{P}_r^\infty(\text{Aut}(F_r)M) > 0$?*

We conjecture that the answer is “no”.

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REFERENCES

- [1] Gilbert Baumslag, *Positive one-relator groups*, Transactions of the American Mathematical Society **156** (May 1971) (en).
- [2] Katalin Bencsáth, Andrew Douglas, and Delaram Kahrobaei, *Some residually solvable one-relator groups*, Irish Mathematical Society Bulletin **0065** (2010), 23–31 (en).
- [3] Emma Dinowitz, Lucy Koch-Hyde, Siobhan O’Connor, and Eamonn Olive, *Growth and language complexity of potentially positive elements of free groups*, arXiv, 2025. arXiv:2512.13967 [math].
- [4] Ralph H. Fox, *Free differential calculus I: Derivation in the free group ring*, The Annals of Mathematics **57** (May 1953), no. 3, 547 (en).
- [5] Richard Goldstein, *An algorithm for potentially positive words in $\$F_2\$$* , Contemporary Mathematics **421** (2006), 157–168 (en).
- [6] Ilya Kapovich, Alexei Myasnikov, and Vladimir Shpilrain, *Combinatorial group theory: Open problems on infinite groups*, De Gruyter, 2026 (en).
- [7] Lucy Koch-Hyde, Siobhan O’Connor, and Eamonn Olive, *Complexity of recognizing potentially positive elements in the free group of rank 2*, International Journal of Algebra and Computation (en).
- [8] H. W. Kuhn, *Subgroup theorems for groups presented by generators and relations*, The Annals of Mathematics **56** (July 1952), no. 1, 22 (en).
- [9] Marco Linton, *Residually rationally solvable one-relator groups*, Bulletin of the London Mathematical Society (October 2025), blms.70210 (en).

- [10] A. Myasnikov, V. Roman'kov, A. Ushakov, and A. Vershik, *The word and geodesic problems in free solvable groups*, Transactions of the American Mathematical Society **362** (April 2010), no. 9, 4655–4682 (en).
- [11] Siobhán O'Connor, *Algorithmic problems in automorphic orbits of free groups*, PhD, 2026 (en).
- [12] C. M. Weinbaum, *On relators and diagrams for groups with one defining relation*, Illinois Journal of Mathematics **16** (June 1972), no. 2 (en).

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